

**Programme: B E – Computer Science and Engineering (AI&ML) &
Computer Science and Engineering (Cyber Security)
Internal Assessment – I**

TERM : CIE-1	COURSE NAME: COA
DATE :30-11-2024 TIME: 9:30 to 10:30AM	COURSE CODE : CI34/CY34
MAX MARKS: 30	PORTIONS : L1-L18 (UNIT 1 and UNIT 2)



Mobile Phones are banned

Instructions to Candidates: Answer any two full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Prove the identity $x+wx^1 = x+ w$ using truth table.	L3	CO1	8
B	Describe the working of clocked SR-flip flop with neat circuit diagram and truth table.	L2	CO1	7
2.a	Simplify the following four variable expression $f(A,B,C,D) = \sum m(3,5,7,8,10,11,12,13)$ using K-map/Boolean algebra.	L3	CO2	5
B	Explain CMOS realization of a NAND gate with circuit diagram and truth table.	L2	CO1	5
C	With a neat diagram explain the data transfer between memory and processor?	L2	CO2	5
3.a	Construct the circuit x-NOR gate using standard basic gates?	L3	CO1	5
B	Write a program that can evaluate the expression $TOTAL = A*B+C+D+E*F$. Assume 1-address mode processor.	L3	CO2	6
C	With an example describe the concepts of Big-endian and Little-endian address assignments?	L2	CO2	4

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create



**Programme: B E – Computer Science and Engineering (AI&ML) &
Computer Science and Engineering (Cyber Security)
Internal Assessment – II**

TERM : 3 rd October 2024 to 25 th Jan 2025	COURSE NAME: Computer Organization and Architecture
DATE : 22-01-2025 TIME: 11:00 to 12:00 NOON	COURSE CODE : CI34/CY34
MAX MARKS: 30	PORTIONS : L20 to L42



Mobile Phones are banned

Instructions to Candidates: **Answer any two full questions.**

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Explain the direct-mapped, set-associative, and fully associative cache mapping techniques. Assume a system with 128 KB of main memory and a cache of 32 KB.	L2	CO4	8
b.	(i) Explain in step by step, how would the system handle multiple interrupts when the processor is executing a particular Interrupt Service Routine (ISR)? Assume nested interrupts are enabled. (ii) Consider a computer system with a 16-bit address bus and a 256-entry interrupt vector table. Each entry in the interrupt vector table holds the address of an interrupt service routine (ISR), and the entries are aligned such that each entry occupies 4 bytes. What is the size (in bytes) of the interrupt vector table?	L3	CO3	7
2.a	Assume 4-bit signed binary numbers. (i) Add -6 to +4 and (ii) Subtract -5 from -7 using 2's complement form of binary numbers..	L3	CO3	7
b	Use Booth's Algorithm to multiply the following two 4-bit signed numbers: A = -3 and B = 6 (i) Represent the numbers in 4-bit 2's complement form. (ii) Step through the Booth's algorithm process and show the intermediate results.	L3	CO4	8
3.a	Explain in detail the working of bus arbitration using the Daisy Chain method. Provide a neat diagram to support your explanation.	L2	CO3	8
b	Discuss the basic organization of micro programmed control unit with neat circuit diagram.	L2	CO5	7

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create



EXAMINATIONS SEPTEMBER/OCTOBER 2020
SUPPLEMENTARY SEMESTER / GRADE IMPROVEMENT/
RE-REGISTERED CANDIDATES

Program	: B.E. : Computer Science and Engineering	Semester	: IV
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CS45	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer any one full question from each unit.

UNIT- I

- Explain connection between the Processor and the Memory. CO1 (06)
 - Explain the concept of stack frame and frame pointer with an example. CO1 (06)
 - Register R1 and R2 of a computer contains the decimal values 1200 and 4600. What is the effective address of the memory operand in each of the following instructions:

i) LOAD 20(R1),R5	ii) MOVE #5000,R5
iii) ADD -(R2),R5	iv) SUBTRACT (R1)+,R5

CO1 (08)
- Explain Pipelining and Superscalar Operation. CO1 (08)
 - Demonstrate the use of Indirect Addressing mode with an example. CO1 (06)
 - Describe different formats of encoding machine instruction for different Addressing modes. CO1 (06)

UNIT- II

- Explain the working of a hardwired control unit with a diagram. CO2 (06)
 - Multiply each of the following pairs of signed 2's complement numbers using booth algorithm and bit pairing of the multiplier (Assume A is the Multiplicand and B is the Multiplier)

i) A=010111 B=110110	ii) A=110011 B=101100.
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CO2 (08)
 - Write the sequence of control steps required for the single bus structure for the following instruction Add (R2),R1. CO2 (06)
- Draw and explain the multi-bus organization of a Processor. CO2 (06)
 - Give an Algorithm for Restoring division method. Illustrate the working of the Algorithm for 8/3 and Represent $(1259.125)_{10}$ in IEEE floating point format. CO2 (08)
 - Illustrate the working of CSA with 45 and 63 as the Multiplicand and Multiplier respectively. CO2 (06)

UNIT- III

- Explain with an example, different types of dependencies. CO3 (06)
 - Explain with an example the Classic Five-Stage Pipeline for a RISC Processor. CO3 (06)

- c) Consider the unpipelined processor in the previous section. Assume that it has a 1 ns clock cycle and that it uses 4 cycles for ALU operations and branches and 5 cycles for memory operations. Assume that the relative frequencies of these operations are 40%, 20%, and 40%, respectively. Suppose that due to clock skew and setup, pipelining the processor adds 0.2 ns of overhead to the clock. Ignoring any latency impact, how much speedup in the instruction execution rate will we gain from a pipeline? CO3 (08)
6. a) Discuss different types of data hazards with examples. CO3 (10)
b) Justify the statement with respect to the pipeline technique: "If there are no stalls, the speedup is equal to the number of pipeline stages". CO3 (05)
c) Explain Dynamic Branch Prediction and Branch-Prediction Buffers. CO3 (05)
- UNIT- IV**
7. a) With a neat diagram describe centralized shared memory architecture. CO4 (08)
b) Discuss Directory-Based Cache Coherence Protocols with an example CO4 (06)
c) Differentiate between: CO4 (06)
i. Write-back and write-through cache.
ii. Multicore and Multiprocessor Architectures.
8. a) With a neat diagram demonstrate write invalidate, cache coherence protocol for a private write-back cache showing the states and state transitions for each block in the cache. CO4 (08)
b) Demonstrate different cache mapping techniques with examples. CO4 (08)
c) With a neat diagram discuss different levels in the memory hierarchy. CO4 (04)
- UNIT- V**
9. a) Explain Interrupt priority schemes. CO5 (06)
b) Define bus arbitration? Explain the different variants of the bus arbitration. CO5 (10)
c) Explain parallel input interface circuit. CO5 (04)
10. a) Explain use of DMA controller in a computer system. CO5 (06)
b) Differentiate between synchronous bus and Asynchronous bus. CO5 (10)
c) Explain USB architecture and its key features. CO5 (04)



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(Approved by AICTE, New Delhi & Govt. of Karnataka)

(Autonomous Institute, Affiliated to VTU)
Accredited by NBA & NAAC with 'A' Grade**SEMESTER END EXAMINATIONS FEBRUARY - MARCH 2021**

Program	: B.E.: Information Science and Engineering	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: IS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT- I

1. a) Briefly discuss the four basic functions and Structural components that enables the performance of a computer. CO1 (08)
- b) Briefly discuss on Multicore, MICS and GPGPUS with its usage in the computer system. CO1 (06)
- c) Suppose that a task makes extensive use of floating-point operations, with 40% of the time consumed by floating-point operations. With a new hardware design, the floating-point module is sped up by a factor of K. Find the overall speedup of the system. CO1 (06)
2. a) Briefly discuss the various techniques that are built into a contemporary processor by the designers to fabricate chips of greater density. CO1 (06)
- b) Discuss the Amdahl's law that provides an insight into the performance of parallel processing systems. CO1 (06)
- c) Briefly discuss the Bus Interconnection scheme with appropriate diagram with all the components involved in the design process. CO1 (08)

UNIT – II

3. a) Discuss the method of accessing units of data among memory along with the three performance parameters of a memory. CO2 (06)
- b) Briefly discuss the three level cache organization in a memory structure. CO2 (06)
- c) Briefly discuss the data organization and formatting on a disk that are capable on reading from or writing to a portion of the platter. CO2 (08)
4. a) Briefly discuss the memory hierarchy with appropriate diagram that are required for better performance of a system. CO2 (07)
- b) Discuss the various categories of major functions or requirements for an I/O module. CO2 (07)
- c) Write short notes on Direct Memory Access, mentioning its importance in the computer system. CO2 (06)

UNIT – III

5. a) Using Quine-McClusky Technique, minimize the Boolean Function $F(A,B,C,D)=\sum(0,2,4,5,8,9,12)$. CO3 (10)

- b) Design a sequential circuit to the sequence 000011010010001000 using positive edge triggered IS flip-flop whose function table is given below CO3 (10)

I	S	$Q_{(t+1)}$
0	0	1
0	1	Q
1	0	Q'
1	1	0

6. a) Design a synchronous counter using JK flip-flops by determining the values of inputs J and K by using the excitation table. CO3 (10)
- b) Describe using a flowchart the concept of the Booth's algorithm for two's complement multiplication. Given $x=0101$ and $y=1010$ in two's complement notation compute the product $p = x * y$ with Booth's algorithm. CO3 (10)

UNIT – IV

7. a) What are the typical elements of a machine instruction? Describe the types of locations can hold source and destination operands. CO4 (06)
- b) Explain the advantages and limitations of each addressing modes. CO4 (07)
- c) Describe the effect of a conditional branch on instruction pipeline operation with a neat timing diagram. What do you mean by branch penalty? CO4 (07)
8. a) Explain different types of operations and operands with examples. CO4 (10)
- b) List and explain different types of registers. CO4 (05)
- c) With a neat diagram, describe the simplified view of a processor, indicating its connection to the rest of the system via the system bus. CO4 (05)

UNIT – V

9. a) List the important characteristics of SMP. Explain the advantages of SMP over uniprocessor. CO5 (08)
- b) Define GPU. Compare CPU and GPU with example. CO5 (06)
- c) Explain different micro-operations. CO5 (06)
10. a) Explain the four states in MESI protocol with neat diagram. CO5 (10)
- b) Describe different methods of clustering with benefits and limitations. CO5 (10)

MAKE UP EXAMINATIONS – JUNE 2023

Program	: B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CY34 / CI34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Differentiate between combinational and sequential circuits. Describe the working of full adder with neat circuit diagram and truth table. CO1 (08)
 - Minimize the following expression using laws of Boolean algebra:
 $(A+B+C)(A+\bar{B}+C)(A+B+\bar{C})$. CO1 (04)
 - Describe the working of JK- flipflop with neat circuit diagram and truth table. CO1 (08)
- Describe the three different classes of computing applications and their characteristics. CO1 (08)
 - With neat circuit diagram and truth table, describe 4:1 multiplexer. CO1 (08)
 - Consider a logic circuit with three inputs A, B, and C. Output Y is high for the following conditions:
 - B and C are true
 - A and C are false
 - A, B, and C are true
 - A, B and C are false.
 Minimize the function Y using Law's of Boolean Algebra. CO1 (04)

UNIT - II

- Describe any Five data transfer instructions used by MIPS architecture. CO2 (06)
 - With the help of circuit diagram discuss First version of the multiplication hardware. CO4 (06)
 - List and explain different addressing modes with example. CO2 (08)
- Multiply 4x3 using multiplication algorithm. CO4 (04)
 - With an Example, explain 2 types of MIPS instruction formats. CO2 (06)
 - With the help of circuit diagram discuss an improved version of the division hardware. CO4 (10)

UNIT - III

- Discuss the direct mapping and associative cache memory mapping functions with suitable examples. CO3 (08)
 - Describe different types of Read only memories. CO3 (06)

- c) Explain the basic structure of the memory hierarchy along with a neat diagram. CO3 (06)
6. a) Illustrate the characteristics of virtual memory. CO3 (10)
b) Discuss how to compute the cache performance. Assume the miss rate of an instruction cache is 3% and the miss rate of the data cache is 5%. If a processor has a CPI of 2 without any memory stalls and the miss penalty is 150 cycles for all misses, determine how much faster a processor would run with a perfect cache that never missed. Assume the frequency of all loads and stores is 45%. CO3 (10)

UNIT- IV

7. a) Explain how instruction fetch cycle works along with a neat diagram and also mention the various registers involved in this. CO4 (08)
b) Describe the Hardwired control unit with a neat diagram. CO4 (06)
c) Differentiate between multiprocessor and multicore processor. CO4 (06)
8. a) Describe the process of fetching and executing instructions, using the Datapath. Explain the control sequences and steps followed for execution of an instruction Add, R3, (R2). CO4 (08)
b) Compare and contrast between CISC and RISC-style processors. CO4 (06)
c) Illustrate the functioning of GPU system architecture. CO4 (06)

UNIT - V

9. a) What do you mean by Multithreading? List the difference between Fine Grain multithreading and Coarse grain multithreading. CO5 (10)
b) Classify parallel computers based on Flynn's Taxonomy. CO5 (10)
10. a) Discuss and list the difference between shared memory multiprocessor and message passing multiprocessors in computer architecture. CO5 (10)
b) Differentiate between CPU and GPU. CO5 (10)

MAKE UP EXAMINATIONS – APRIL 2025

Program	: B.E. – CSE (Cyber Security)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CY34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Simplify the following four variable expression CO1 (08)
 $Y = \sum m(0,2,6,7,8,15)$
 *Note: Student can use K-Map or Boolean Algebra approach.
 - List the difference between combinational circuits and Sequential circuits. With the help of Truth -Table design 4: 1 Mux circuit. CO1 (06)
 - Explain clocked JK flip-flop with neat circuit diagram. CO1 (06)
- Simplify the following three variable expression using Boolean algebra. CO1 (08)
 $Y = \sum m(1,2,3)$ and $Y = \prod M(0,2,6,7)$
 - Implement Boolean expression for Ex-OR gate using NOR gates. CO1 (06)
 - List the basic types of shift registers in terms of the data movement and explain any two. CO1 (06)

UNIT - II

- With a neat diagram explain the indirect addressing mode. Write a sequence of machine instruction to add N elements stored in the memory using the same. CO2 (08)
 - Explain the basic performance equation and its parameters. Explain the different ways of optimizing these parameters to improve the performance of a computer. CO2 (06)
 - What are the steps involved in sequencing instructions during the fetch-decode-execute cycle? CO2 (06)
- Demonstrate how the operation $P = (Q * R - S / T) + U * V$ can be implemented in a CO2 (08)
 - One address
 - Two address and
 - Three address instructions
 - What is subroutine linkage method? Explain the execution effect of call and return instructions with respect to the subroutine nesting. CO2 (06)
 - Illustrate the working of single bus organization of a processor with a neat diagram. CO2 (06)

UNIT - III

- With the help of a neat diagram, describe how multiple devices are connected in a daisy chain configuration to handle interrupts. Explain how the priority of devices is determined and how an interrupt request is passed along the chain. CO3 (08)

- b) Compare synchronous buses and asynchronous buses in a computer system in terms of speed, complexity, scalability, and cost. CO3 (06)
- c) In a binary representation, a 4-bit number value is 1111. What is the value it represents in (I) sign-magnitude, (II) 1's complement, and (III) 2's complement. CO3 (06)
- 6. a) With neat diagram(s), describe the steps involved in a DMA data transfer between an I/O device and memory. CO3 (08)
- b) Compare and contrast DMA-controlled and interrupt-driven methods for transferring data from an input device to memory. CO3 (06)
- c) Design a 4-bit carry lookahead adder. CO3 (06)

UNIT- IV

- 7. a) What are the main differences between Semiconductor RAM (SRAM and DRAM) and ROM (Read-Only Memory)? Discuss their respective uses in a computer system. CO4 (08)
- b) Explain the concept of cache mapping functions. Describe the Direct Mapping technique in cache memory with an example. CO4 (06)
- c) Explain the basic concept of cache memory and its role in improving system performance. CO4 (06)
- 8. a) Discuss cache replacement algorithms (FIFO, LRU, and Optimal). Explain how the choice of a replacement algorithm affects the performance of the cache memory system. CO4 (07)
- b) Explain the trade-offs between speed, size, and cost in the design of memory systems. CO4 (07)
- c) Explain the process of multiplying two positive integers using the standard multiplication algorithm. CO4 (06)

UNIT – V

- 9. a) List out the key aspects of GPU programs driving GPU processor architecture and explain any one in briefly. CO5 (10)
- b) Discuss the performance of parallel computers based on Amdahl's Law. CO5 (10)
- 10. a) List out the difference of CISC and RISC style Processors. CO5 (08)
- b) Discuss the operation of four stage pipeline. CO5 (06)
- c) Illustrate along with key characteristic as how GPUs vary from CPUs. CO5 (06)

SEMESTER END EXAMINATIONS – APRIL 2023

Program	: B.E. - CSE (Cyber Security) / CSE (Artificial Intelligence and Machine Learning)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CY34 / CI34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Simplify the following three variable expression using Boolean algebra $Y = \sum m(0,2,6,7)$. CO1 (04)
 - Design half adder and Full adder circuit. CO1 (06)
 - Briefly describe the working of clocked RS flip-flop with circuit diagram and truth table. CO1 (05)
 - Describe the working of 4 bit asynchronous counter with a neat diagram. CO1 (05)
- Simplify the following three variable expression using Boolean algebra $Y = \sum m(1,5,7)$ and $Y = \prod M(0,2,6,7)$. CO1 (06)
 - With neat circuit diagram describe half Subtractor and Full Subtractor. CO1 (06)
 - Define race condition? With the help of Truth table Explain JK flip-flop. CO1 (08)

UNIT - II

- Perform sequential multiplication on (10×15) using version-1 multiplication algorithm. CO2 (06)
 - Describe any two types of MIPS instruction formats with an example for each type. CO2 (06)
 - With an example, explain the different ways in which the operands can be specified in an assembly language instruction. CO2 (08)
- Perform binary division on the following using improved division algorithm.
Dividend=15 and Divisor=4. CO2 (06)
 - Describe the two types of MIPS instruction formats. In MIPS, if registers \$S0 to \$S7 map to 16 to 23 and registers \$t0 to \$t7 map on to 8 to 15. Add instruction is identified by (0,32) and lw instruction is identified as 35. Convert the following MIPS assembly instruction to machine code. CO2 (06)
 - Add \$t0, \$S1, \$S2
 - lw \$S1, 100(\$S2)
 - Give the IEEE 754 binary representation of the number -0.75_{10} in single precision. CO2 (08)

UNIT - III

5. a) With a neat diagram describe the read and write operation on a single DRAM cell. CO3 (10)
- b) Consider a computer system with main memory of size 128 bytes and cache memory of size 32 bytes divided in to 8 bytes partition. Explain different cache mapping techniques to map blocks of main memory to cache lines. CO3 (10)
6. a) Give the memory hierarchy of the various storage components of a computer in the order of increasing speed and cost per bit. CO3 (05)
- b) Describe the following terms: CO3 (10)
- i. Spatial locality
 - ii. Temporal locality
 - iii. Cache hit
 - iv. Cache Miss
 - v. Write through protocol.
- c) Consider a direct mapped cache of size 16 KB with block size 256 bytes. The size of main memory is 128 KB. Find CO3 (05)
- i. Number of bits in tag
 - ii. Tag directory size.

UNIT- IV

7. a) With neat diagram explain the GPU System Architecture. CO4 (10)
- b) List out the difference of CISC and RISC style Processors. CO4 (06)
- c) Describe CUDA used in GPU's. CO4 (04)
8. a) List out the key aspects of GPU programs driving GPU processor architecture and explain any one in detail. CO4 (10)
- b) Differentiate between multiprocessor and multicore processor. List out the benefits of multiprocessor systems over uniprocessor systems. CO4 (10)

UNIT – V

9. a) Discuss the various difficulties associated with creation of parallel processing programs. CO5 (06)
- b) Discuss the Flynn's classification of parallel processing systems. CO5 (08)
- c) What is threading? Explain the hardware multithreading in detail. CO5 (06)
10. a) Discuss the Classic organization of a shared memory multiprocessor. CO5 (06)
- b) Illustrate along with key characteristics as how GPUs vary from CPUs. CO5 (06)
- c) Explain the different network topologies followed in multiprocessor systems. CO5 (08)

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS - FEB / MARCH 2025

B.E. - CSE (Cyber Security) /			
Program	: CSE (Artificial Intelligence and Machine Learning)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CY34 / CI34 (2021,2022,2023 Batch)	Duration	: 3 Hrs.

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Using the basic theorems and postulates of Boolean algebra, simplify the following Boolean expression: $F = x'y'z + xyz + x'yz + xy'z$. CO1 (06)
 - Explain how a NAND gate can be used to synthesize any other basic logic gates. Demonstrate how to implement an AND, OR, and NOT gate using only NAND gates. CO1 (07)
 - What is a multiplexer? Design a 4-to-1 multiplexer and explain its working with a block diagram and truth table. CO1 (07)
- Design a 3-to-8 decoder and explain how it works with a block diagram and truth table. CO1 (08)
 - Design half subtractor and full subtractor. CO1 (06)
 - Explain the operation of an RS flip-flop, including its circuit diagram and truth table. CO1 (06)

UNIT - II

- Explain any Five data transfer instructions used by x86 architecture. CO2 (06)
 - List the difference between high-level language and machine language? CO2 (06)
 - Discuss Big endian with little endian with respect to byte addressability with an example. CO2 (08)
- With an example discuss the direct and indirect addressing modes. CO2 (06)
 - Explain assembly directive? List the few assembly directives and which directive are used to inform the assembler. CO2 (06)
 - With Example, explain the different ways in which the operands can be specified in an assembly language instruction. CO2 (08)

UNIT - III

- Illustrate the 4-bit carry-lookahead adder with a neat diagram. Explain how does it differs from a ripple-carry adder. CO3 (08)
 - Explain the difference between programmed I/O and interrupt-driven I/O. CO3 (07)
 - Compare the SCSI bus interface with the PCI bus in terms of functionality and use cases. CO3 (05)

6. a) Explain direct memory access (DMA), and how does it improve I/O performance? CO3 (06)
- b) Convert the following pairs of decimal numbers to 6 bit signed, 2's complement binary numbers and perform the addition and subtraction between them. State whether overflow occurs or not. CO3 (08)
- i) -14 and -9
 - ii) 6 and 11
 - iii) -13 and 18
 - iv) 15 and -10
- c) Describe how the SCSI bus facilitates communication with storage devices. CO3 (06)

UNIT- IV

7. a) Discuss the role of cache memory, main memory (RAM), and secondary storage in the memory hierarchy. CO4 (06)
- b) Compare the trade-offs between speed, cost, and capacity for each level of the hierarchy. CO4 (06)
- c) Given the following 4-bit binary numbers: $A=1101_2$ and $B=0110_2$. Multiply A and B using the standard binary multiplication method. Show the intermediate steps of the multiplication. CO4 (08)
8. a) Explain the working principles of FIFO (First-In-First-Out), LRU (Least Recently Used), and Random Replacement algorithms. Provide an example scenario where each algorithm might be most suitable. CO4 (08)
- b) Why is DRAM preferred for main memory, while SRAM is used for cache memory? CO4 (04)
- c) Perform the binary division of 110110_2 (dividend) by 101_2 (divisor) using the long division method. Show the step-by-step process of the division, including subtraction steps and how the quotient and remainder are determined. CO4 (08)

UNIT - V

9. a) What are the basic functions of a processing unit? Explain the role of the Control Unit in executing instructions. CO5 (07)
- b) Explain the difference between hard-wired control and microprogrammed control in the context of control units. CO5 (06)
- c) Describe the concept of multiple bus organization in a computer system. How does it improve the performance of instruction execution? CO5 (07)
- 10 a) Discuss the basic concepts of pipelining in processors. Explain the working of a RISC instruction pipeline and how it enhances performance. CO5 (08)
- b) Explain how to improve the performance in a pipelined system and discuss its limitations. CO5 (06)
- c) What are pipeline hazards? Explain the three types of hazards (structural, data, and control hazards). CO5 (06)

SEMESTER END EXAMINATIONS – MARCH 2024

Program	: B.E. – CSE (Artificial Intelligence and Machine Learning) / CSE (Cyber Security)	Semester	: III
Course Name	: Computer Organization and Architecture	Max. Marks	: 100
Course Code	: CI34 / CY34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

- Differentiate between combinational and sequential circuits. Describe the working of full adder with neat circuit diagram and truth table. CO1 (08)
 - Minimize the following expression using laws of Boolean algebra:
 $(A+B+C) (A+\bar{B}+C) (A+B+\bar{C})$. CO1 (04)
 - Describe the working of JK- flipflop with neat circuit diagram and truth table. CO1 (08)
- Describe the working of RS flipflop with the neat circuit diagram and truth table. CO1 (08)
 - With neat circuit diagram and truth table briefly describe 4:1 multiplexer. CO1 (08)
 - Consider a logic circuit with three inputs A, B, C. Output Y is high for the following conditions:
 - B and C are true
 - A and C are false
 - A, B, and C are true
 - A, B and C are false.

Minimize the function Y using Law's of Boolean Algebra.

UNIT – II

- Explain any Five data transfer instructions used by MIPS/x86 architecture. CO2 (06)
 - List the difference between high-level language and machine language? CO2 (06)
 - How to measure the performance of computer? Explain. CO2 (08)
- With an example discuss the relative addressing mode. CO2 (06)
 - Explain assembly directive? List the few assembly directives and which directive are used to inform the assembler. CO2 (06)
 - With an Example, explain Direct and indirection instruction set of MIPS/x86 assembly language. CO2 (08)

UNIT – III

- Explain with a neat diagram bus arbitration using Daisy chain. CO3 (08)
 - Explain with a neat diagram how the DMA controllers are used in computer system. CO3 (06)
 - Design and Implement 4 bit carry look ahead adder with neat circuit diagram. CO3 (06)

6. a) Explain the I/O interface for an input device. CO3 (06)
b) List the functions of interface circuits and describe how the Printer to processor connection is handled in a system with a neat block diagram. CO3 (08)
c) Explain Binary addition- subtraction logic network. CO3 (06)

UNIT – IV

7. a) Describe the following: CO3 (06)
i. Memory access time
ii. Memory cycle time
iii. Function of memory management unit
iv. Logical or virtual memory
v. Significance of cache memory.
b) Illustrate the working of static random access and dynamic random-access memories with their block diagrams. CO3 (06)
c) Write the algorithm of restoring division method. Perform the division on the numbers 16/4 using the same algorithm and write register circuit arrangement to perform the same. CO3 (08)
8. a) Illustrate associative and set-associative cache memory mapping techniques with an example. CO3 (06)
b) Perform the multiplication on the numbers 10 x 08 using ripple carry addition method and carry save addition method. Write the carry-save adder circuit diagram for 4-bit operands and mention reasons why CSA working faster than the ripple carry adder. CO3 (08)
c) Perform multiplication on the numbers 8 x -11 using Booth recoding and Booth bit pair coding methods. CO3 (06)

UNIT- V

9. a) List out the key aspects of GPU programs driving GPU processor architecture and explain any one in briefly. CO5 (06)
b) List out the difference of CSIC and RISC style processors. CO5 (08)
c) Discuss the basic organization of micro programmed control unit with neat circuit diagram. CO5 (06)
10. a) List the sequence of actions needed to fetch and execute the instruction Store R6, X(R8) and Load R5, X(R7). CO5 (06)
b) Illustrate along with key characteristics as how GPUs vary from CPUs. CO5 (08)
c) Explain the working of Hardwired Control Unit with a neat diagram. CO5 (06)
